LECTURE 7

Capacitors

Capacitor definition

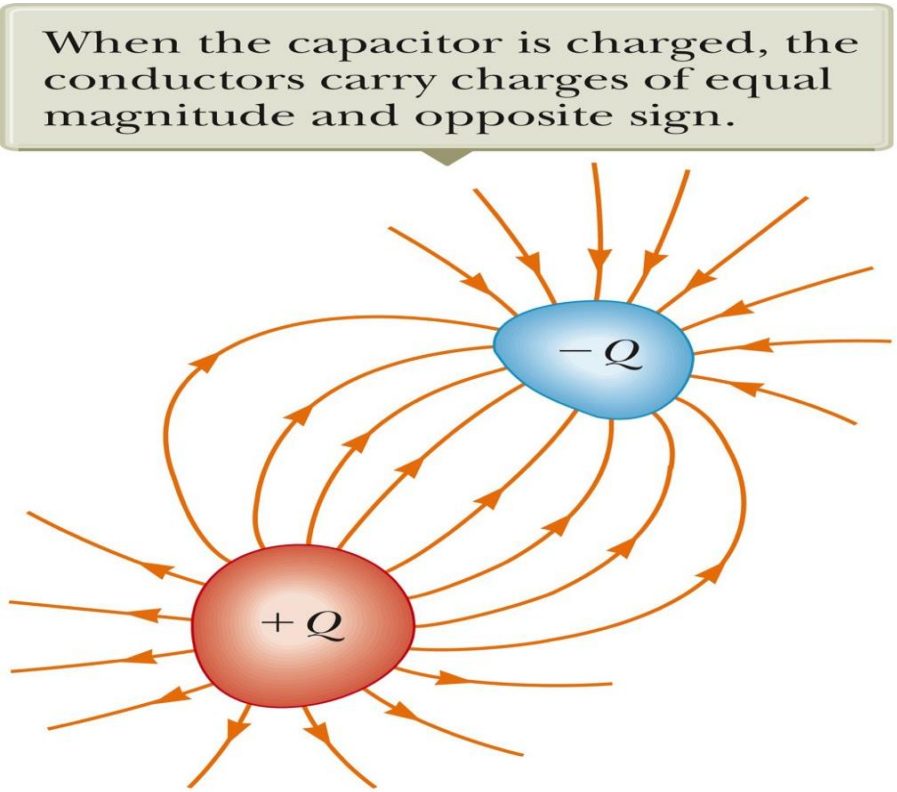
Capacitors are devices that store electric charge.

Examples of where capacitors are used include:

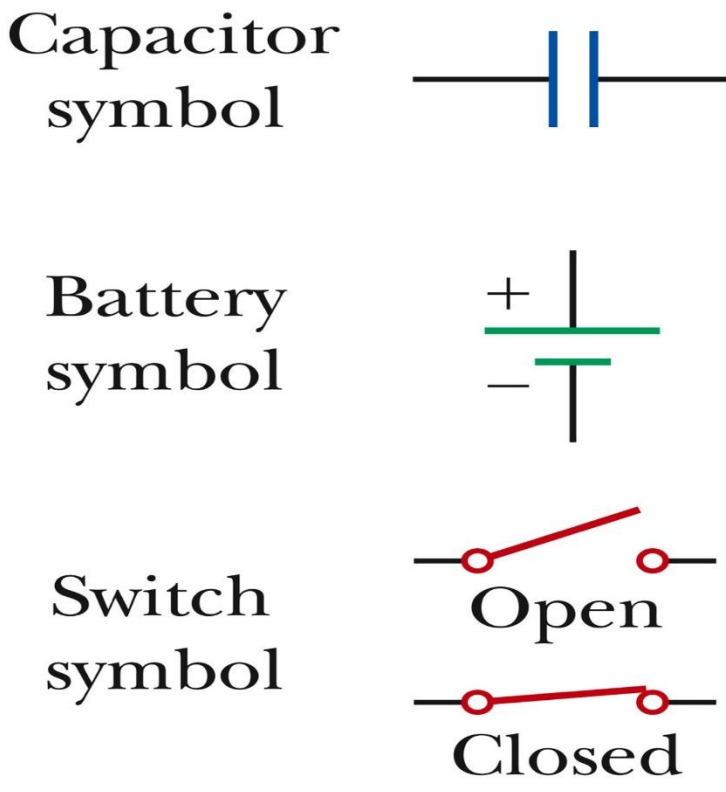
- radio receivers
- filters in power supplies
- energy-storing devices in electronic flashes



- A capacitor consists of two conductors.
- These conductors are called plates.
- When the conductor is charged, the plates carry charges of equal magnitude and opposite directions.
- A potential difference exists between the plates due to the charge.



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Definition of Capacitance

The **capacitance**, C, of a capacitor is defined as the ratio of the magnitude of the charge on either conductor to the potential difference between the conductors.

$$C \equiv \frac{Q}{\Delta V}$$

- The capacitance is a measure of the capacitor's ability to store charge .
- The capacitance of a capacitor is the amount of charge the capacitor can store per unit of potential difference.
- The SI unit of capacitance is the **farad** (F).
- The farad is a large unit, typically you will see microfarads (μF) and picofarads (pF).

Capacitance

$$C = \frac{Q}{\Delta V}$$

Parallel Plates

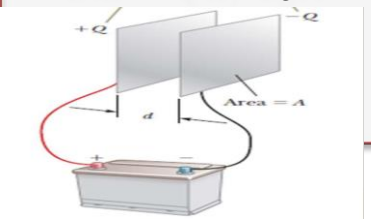
The charge density on the plates is

$\sigma = Q/A.$

The capacitance is proportional to the area of its plates and inversely proportional to the distance between the plates.

$$C = \frac{Q}{\Delta V}$$

$$C = \frac{Q}{E.d} = \frac{Q}{Qd/\varepsilon_o A} = \frac{\varepsilon_o A}{d}$$

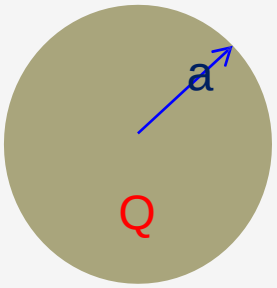


Isolated Sphere

Assume a spherical charged conductor with radius *a.*

$$C = \frac{Q}{\Delta V}$$

$$= \frac{Q}{k_e Q/a} = \frac{R}{k_e} = 4\pi\varepsilon_o a$$



a Cylindrical Capacitor

$\Delta V = -2k_e\lambda \ln(b/a)$

$\lambda = Q/\ell$

The capacitance is

$$C = \frac{Q}{\Delta V}$$

$$= \frac{\ell}{2k_e \ln(b/a)}$$



a Spherical Capacitor

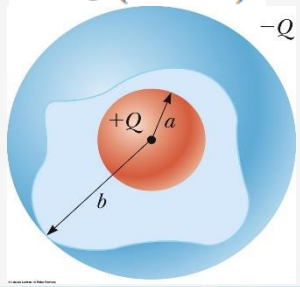
The potential difference will be

$$\Delta V = k_e Q\left(\frac{1}{b} - \frac{1}{a}\right)$$

The capacitance will be

$$C = \frac{Q}{\Delta V}$$

$$= \frac{ab}{k_e(b-a)}$$



Example 1

A parallel-plate capacitor with air between the plates has an area $A = 2.00X10^{-4}\ m^2$ and a plate separation d=1.00 mm. Find its capacitance.

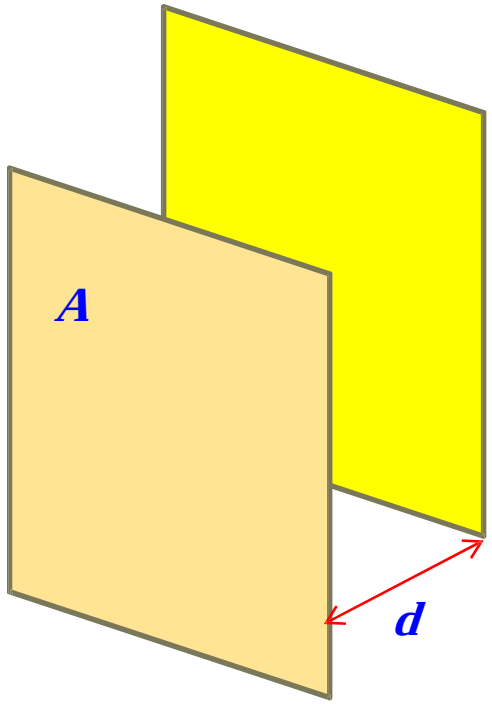
Solution:

$$C = \frac{\varepsilon_o A}{d}$$

$$C = \frac{(8.85 \times 10^{-12})(2 \times 10^{-4})}{1 \times 10^{-3}}$$

$$C = 1.77 \times 10^{-12} F$$

$$C = 1.77\ pF$$



Energy Stored in a Capacitor

Assume the capacitor is being charged and, at some point, has a charge q on it.

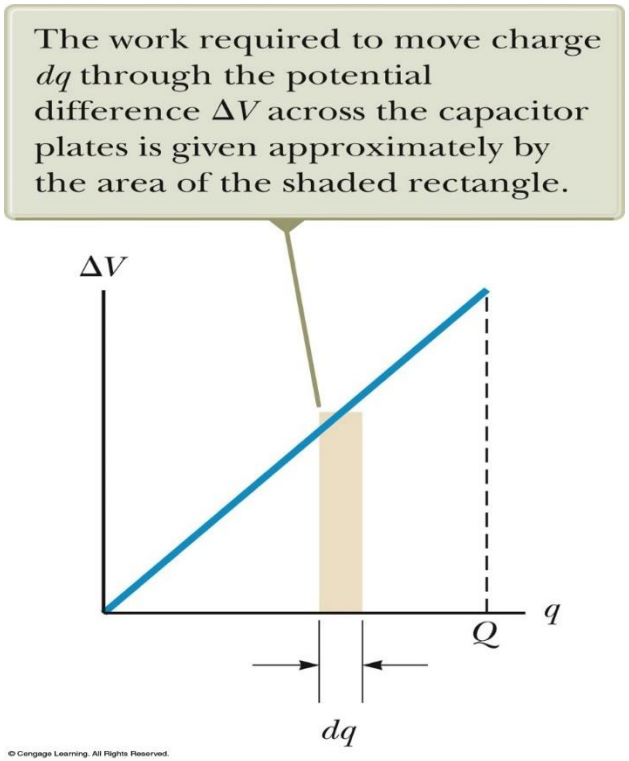
The work needed to transfer a charge from one plate to the other is

$$dW = \Delta V dq = \frac{q}{C} dq$$

The work required is the area of the tan rectangle.

The total work required is

$$W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}$$



The work done in charging the capacitor appears as electric potential energy U

$$U = \frac{Q^2}{2C} = \frac{1}{2}Q\Delta V = \frac{1}{2}C(\Delta V)^2$$

- This applies to a capacitor of any geometry.
- The energy stored increases as the charge increases and as the potential difference increases.
- In practice, there is a maximum voltage before discharge occurs between the plates.
- The energy can be considered to be stored in the electric field .
- For a parallel-plate capacitor, the energy can be expressed in terms of the field as $U = \frac{1}{2} (\varepsilon_o Ad)E^2$
- It can also be expressed in terms of the energy density (energy per unit volume)

$$U_E = \frac{1}{2} \varepsilon_o E^2.$$

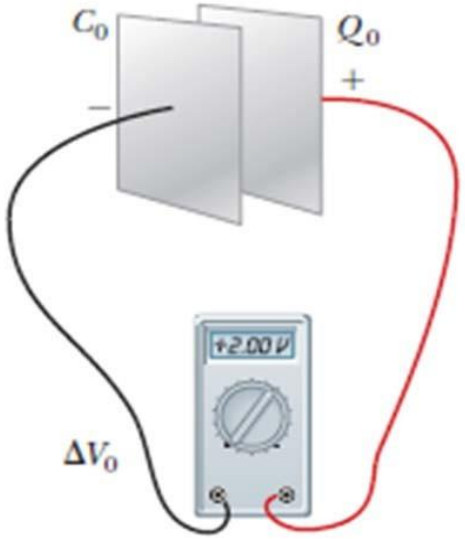
Capacitors with Dielectrics

A dielectric is a nonconducting material that, when placed between the plates of a capacitor, increases the capacitance.

such as rubber, glass, and waxed paper

Consider parallel-plate capacitor that without a dielectric has a charge Q_0 and a capacitance C_0. The potential difference across the capacitor is

$$V_o = \frac{Q_o}{C_o}$$

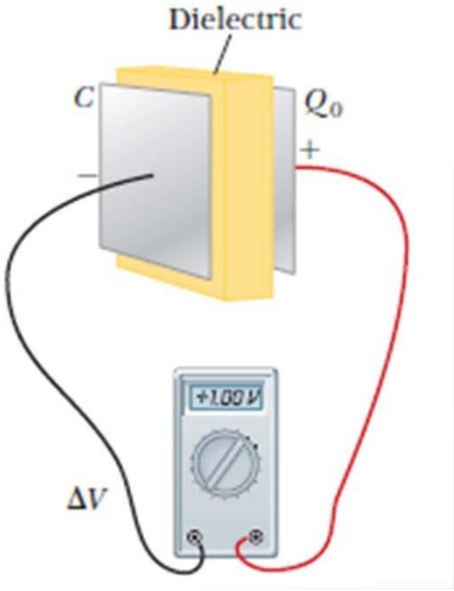


If a dielectric is now inserted between the plates as in Figure, the voltmeter indicates that the voltage between the plates decreases to a value V. The voltages with and without the dielectric are related by a factor k as follows:

$$V = \frac{V_o}{K}$$

K is dielectric constant

$$C = \frac{Q_o}{V} = \frac{Q_o}{V_o/K} = K\frac{Q_o}{V_o}$$



With a dielectric, the capacitance becomes

$$C = KC_o$$

$$C = K\varepsilon_o\frac{A}{d}$$

The capacitance increases by the factor κ when the dielectric completely fills the region between the plates.

Some Dielectric Constants and Dielectric Strengths :

Table 26.1 **Approximate Dielectric Constants and Dielectric Strengths of Various Materials at Room Temperature**

Material	Dielectric Constant κ	Dielectric Strength[a] (10^6 V/m)
Air (dry)	1.000 59	3
Bakelite	4.9	24
Fused quartz	3.78	8
Mylar	3.2	7
Neoprene rubber	6.7	12
Nylon	3.4	14
Paper	3.7	16
Paraffin-impregnated paper	3.5	11
Polystyrene	2.56	24
Polyvinyl chloride	3.4	40
Porcelain	6	12
Pyrex glass	5.6	14
Silicone oil	2.5	15
Strontium titanate	233	8
Teflon	2.1	60
Vacuum	1.000 00	—
Water	80	—

[a]The dielectric strength equals the maximum electric field that can exist in a dielectric without electrical breakdown. These values depend strongly on the presence of impurities and flaws in the materials.

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Example 2

A capacitor has plates of area 0.034m² and a separation of 2mm in air. The potential difference between the plates is 200 V.

a) Calculate the capacitance.

b) Calculate the electric field between the plates.

c) Calculate the charge on each plate.

Solution

a) Calculate the capacitance.

$$C = \frac{\varepsilon_o A}{d} \qquad C = \frac{(8.85 \times 10^{-12})(0.034)}{2 \times 10^{-3}} = 150.45 \times 10^{-12} F = 150.45 pF$$

b) Calculate the electric field between the plates.

$$E = \frac{V}{d} = \frac{200}{2 \times 10^{-3}} = 10^5 volt/m$$

c) Calculate the charge on each plate.

$$Q = CV$$

$$Q = (150.45 \times 10^{-12})(200)$$

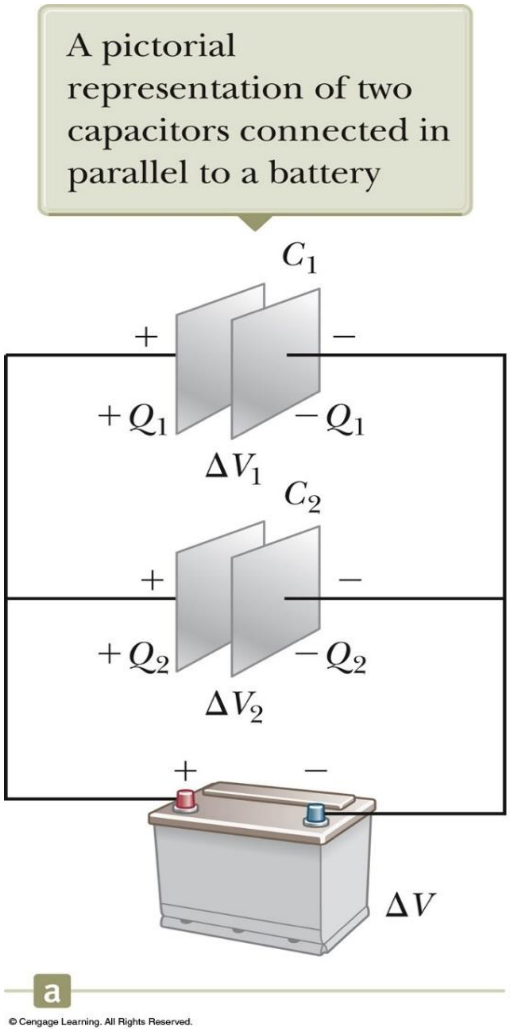
$$Q = 30.1 \times 10^{-9} C$$

Capacitors in series and parallel

Capacitors in Parallel

When capacitors are first connected in the circuit, electrons are transferred from the left plates through the battery to the right plate, leaving the left plate positively charged and the right plate negatively charged.

The flow of charges ceases when the voltage across the capacitors equals that of the battery.



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The potential difference across the capacitors is the same. And each is equal to the voltage of the battery

$$\Delta V_1 = \Delta V_2 = \Delta V$$

** ΔV is the battery terminal voltage

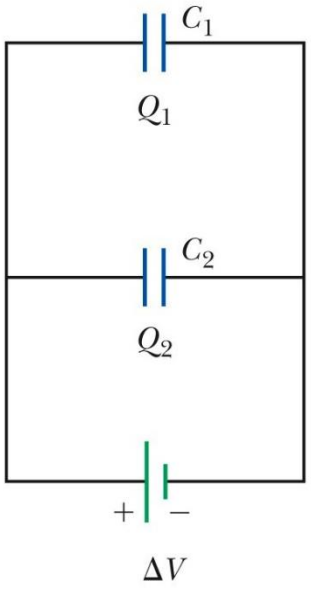
The total charge is equal to the sum of the charges on the capacitors.

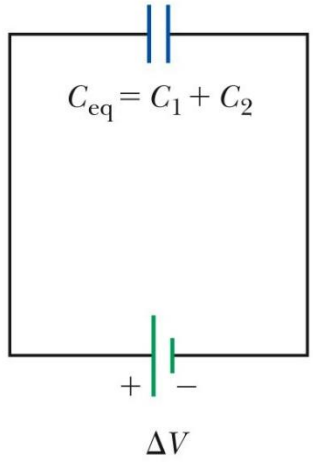
$$Q_{tot} = Q_1 + Q_2$$

The capacitors reach their maximum charge when the flow of charge ceases.

A circuit diagram showing the two capacitors connected in parallel to a battery

A circuit diagram showing the equivalent capacitance of the capacitors in parallel





b c

- The capacitors can be replaced with one capacitor with a capacitance of C_{eq}.

- The equivalent capacitor must have exactly the same external effect on the circuit as the original capacitors.

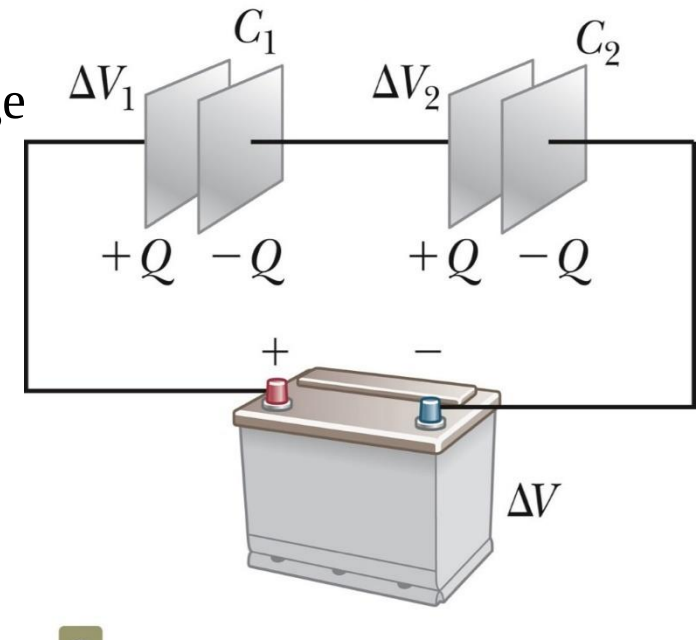
- $C_{eq} = C_1 + C_2 + C_3 + \ldots$

- The equivalent capacitance of a parallel combination of capacitors is greater than any of the individual capacitors.

Capacitors in Series

- When a battery is connected to the circuit, electrons are transferred from the left plate of C_1 to the right plate of C_2 through the battery.
- As this negative charge accumulates on the right plate of C_2, an equivalent amount of negative charge is removed from the left plate of C_2, leaving it with an excess positive charge.
- All of the right plates gain charges of –Q and all the left plates have charges of +Q.

A pictorial representation of two capacitors connected in series to a battery



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The potential differences add up to the battery voltage.

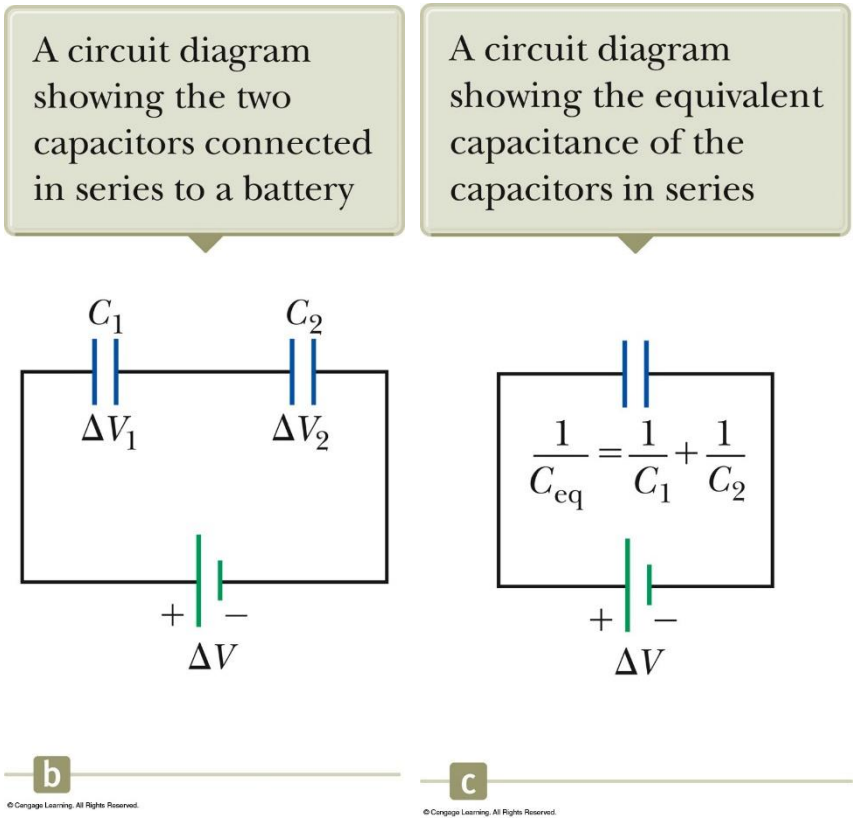
$$\Delta V_{tot} = \Delta V_1 + \Delta V_2 + \ldots$$

The charges are all the same.

$$Q_1 = Q_2 = Q$$

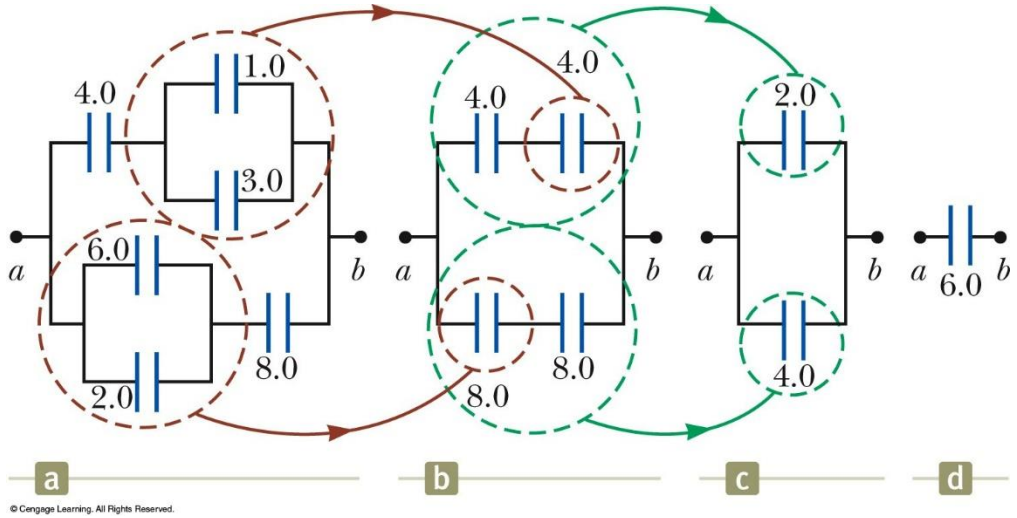
The equivalent capacitance is

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots$$



* The equivalent capacitance of a series combination is always less than any individual capacitor in the combination.

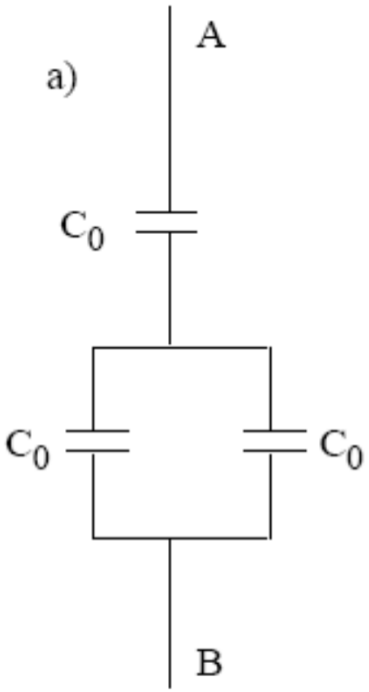
Equivalent Capacitance, Example

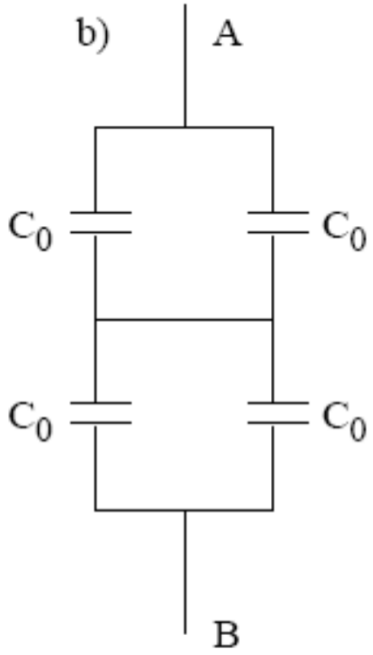


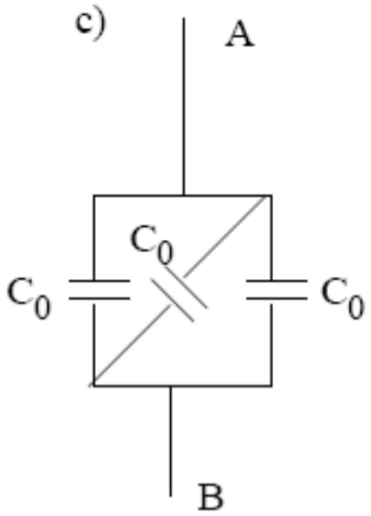
- The 1.0-mF and 3.0-mF capacitors are in parallel as are the 6.0-mF and 2.0-mF capacitors.
- These parallel combinations are in series with the capacitors next to them.
- The series combinations are in parallel and the final equivalent capacitance can be found.

Example 1

Determine the capacitance of each of the networks shown in the figure.

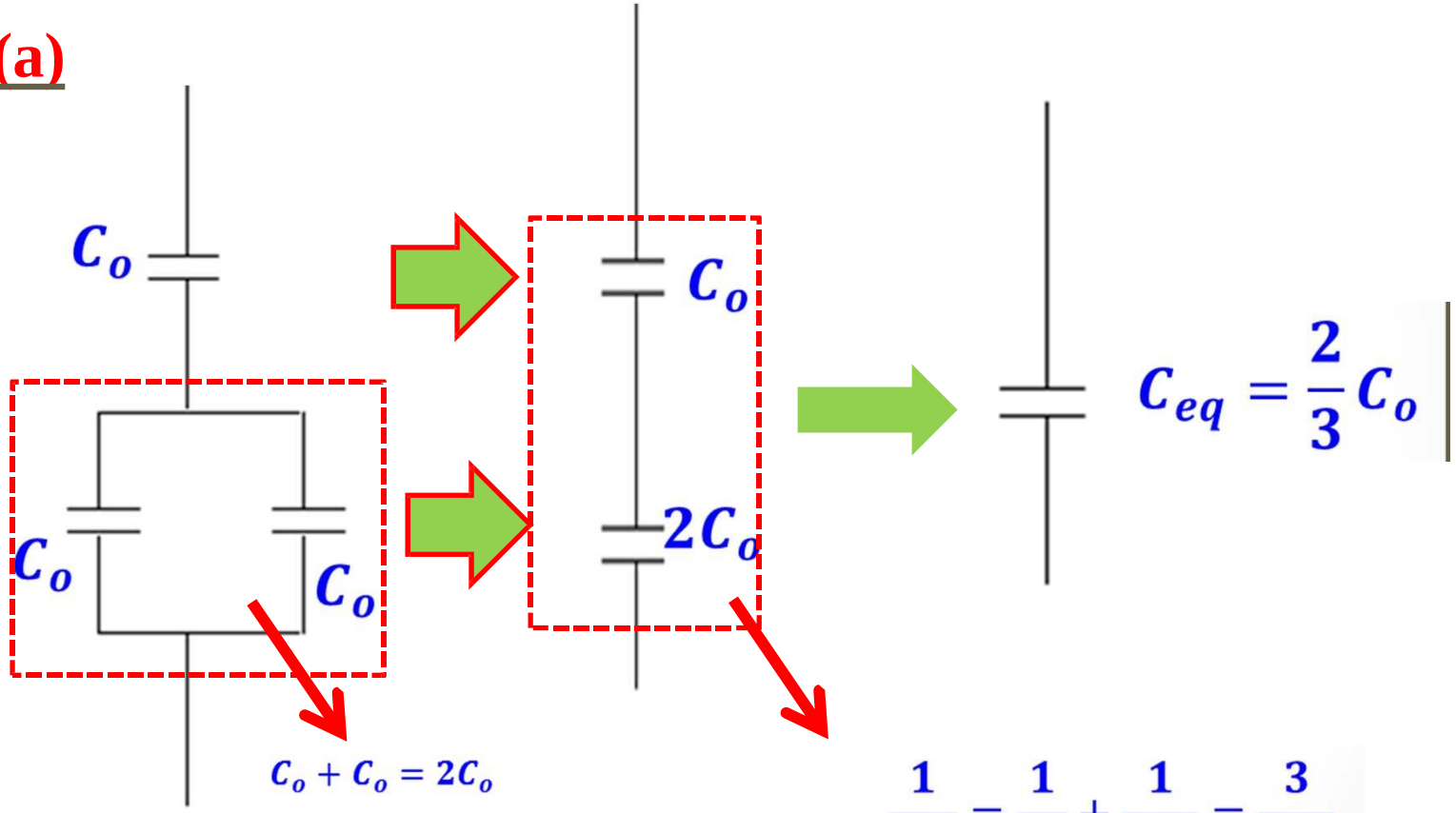






Solution

Fig. (a)



$$\frac{1}{C_{eq}} = \frac{1}{C_o} + \frac{1}{2C_o} = \frac{3}{2C_o}$$

$$C_{eq} = \frac{2}{3}C_o$$

Fig. (b)

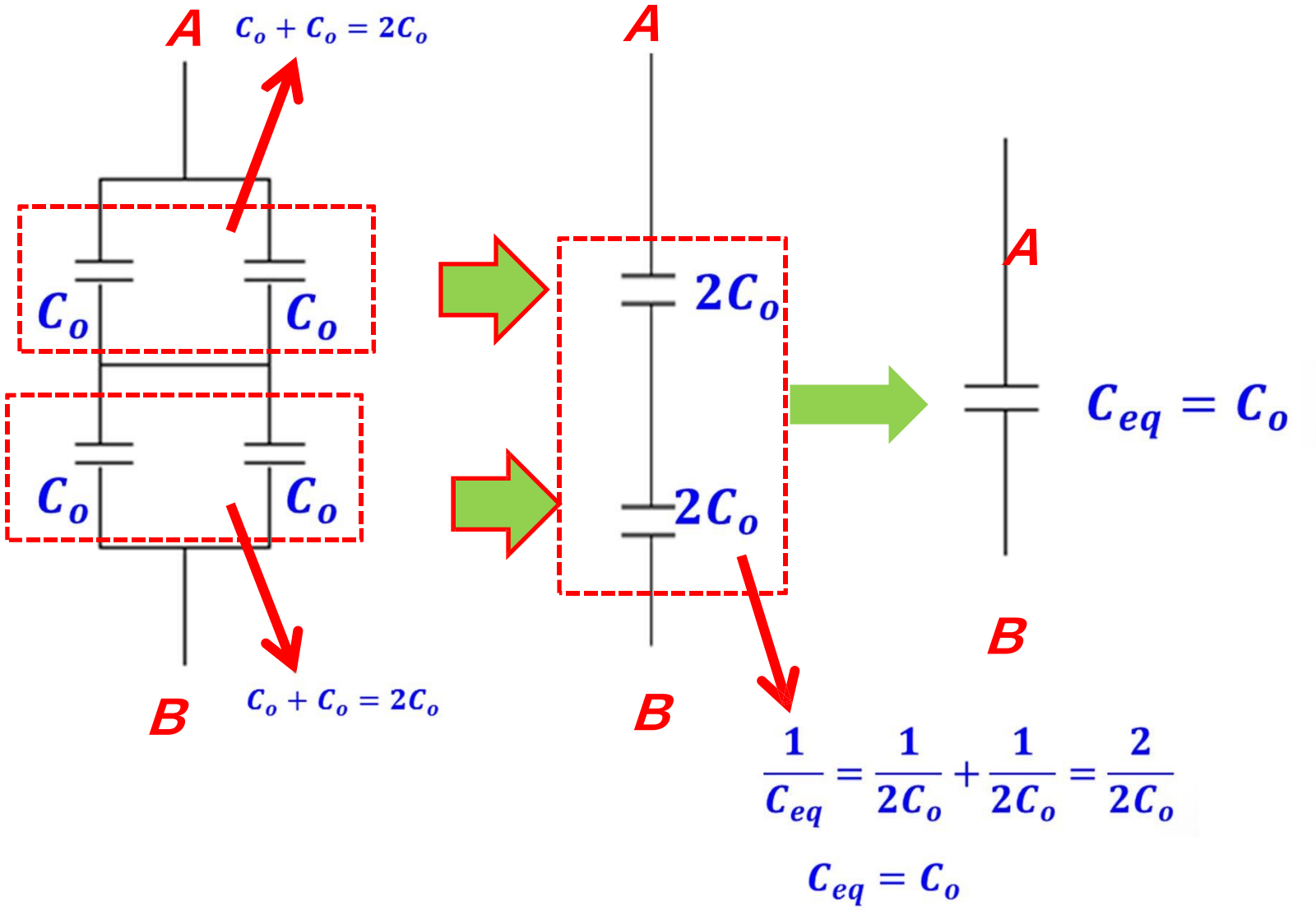
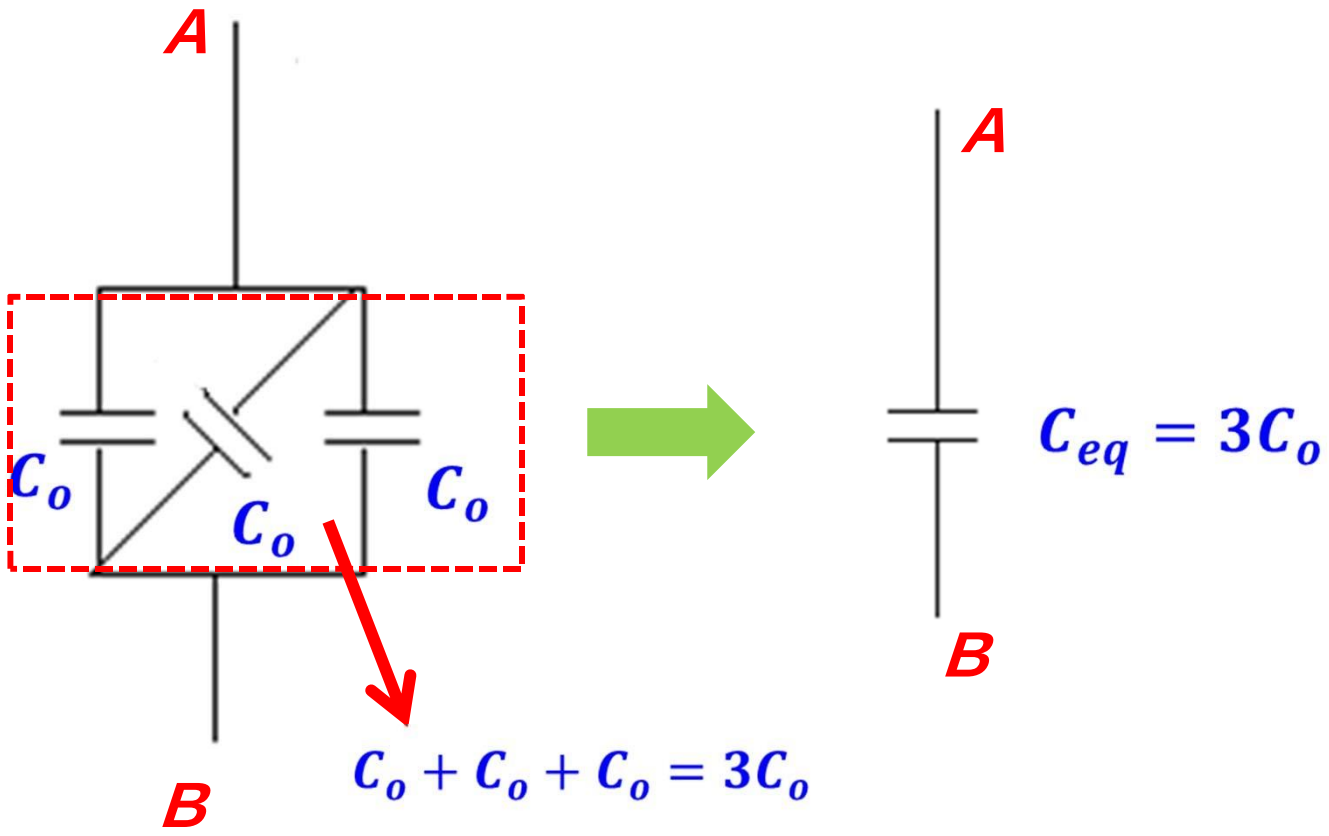

A
$C_o + C_o = 2C_o$
C_o
C_o
C_o
C_o
B
$C_o + C_o = 2C_o$
A
$2C_o$
$2C_o$
B
A
$C_{eq} = C_o$
B
$\frac{1}{C_{eq}} = \frac{1}{2C_o} + \frac{1}{2C_o} = \frac{2}{2C_o}$
$C_{eq} = C_o$

Fig. (c)

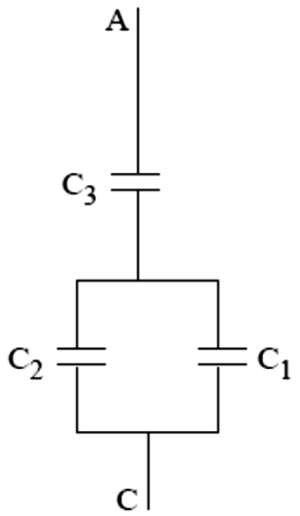


Example 2

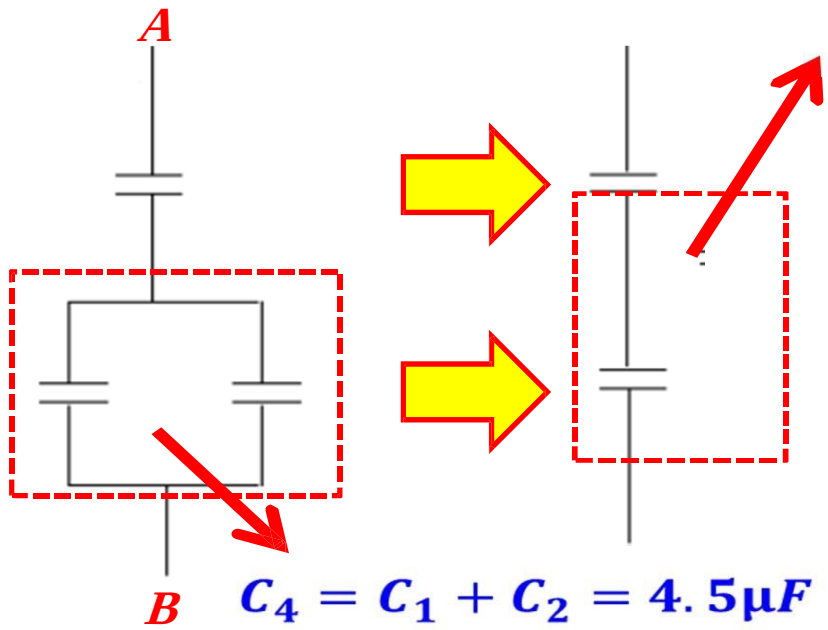
In the following circuit of capacitors, where $C_1 = 3\ \mu F$, $C_2 = 1.5\ \mu F$ and $C3 = 2\ \mu F$

a) Find the equivalent capacitance between AC.

b)If a battery of 12 V is connected between A and C, find the total charge on the capacitors C_1, C_2 and C_3.

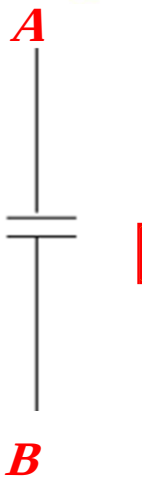


Solution



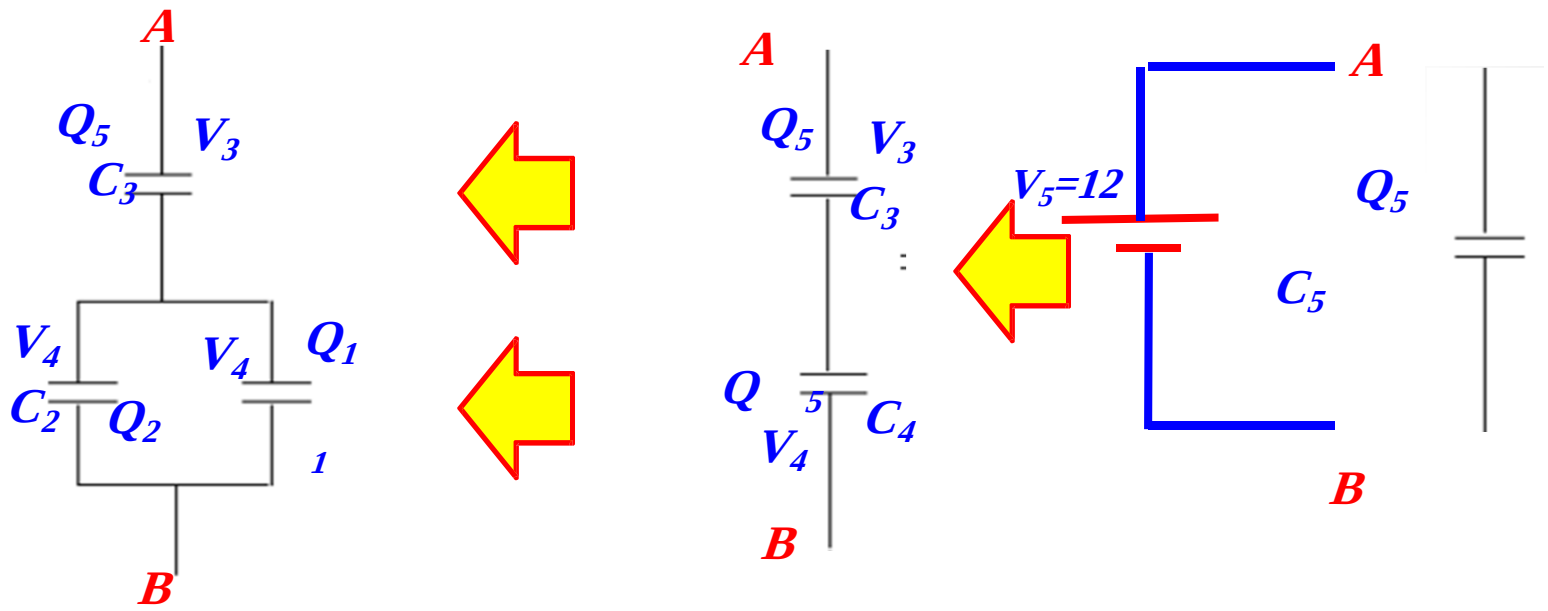
$C_4 = C_1 + C_2 = 4.5\mu F$

$$\frac{1}{C_5} = \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{2} + \frac{1}{4.5}$$



$$C_5 = 1.385\mu F$$

(b) If a battery of 12 V is connected between A and B, find the total charge on the capacitors C_1, C_2 and C_3.



The charge on C_5 is

$$Q_5 = C_5 V = 1.3846 \times 12 = 16.61 \mu C$$

Q_5 is the same charge on C_3 and C_4,

$$Q_3 = Q_4 = Q_5 = 16.61 \mu C$$

The potential a cross C

$$V_4 = \frac{Q_4}{C_4} = \frac{16.61}{4.5} = 3.7 volt$$


Thank
You